

RAG Chapter 1: Complete Code-to-Dataset Mapping

A complete, 41-script execution guide mapping every Python code file in 01-DataLoading/ to its GitHub sample dataset and detailing the exact output produced.

1. Simple Text & Directory Loading (01-SimpleTextLoading/)

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
01-LoadTxtFileWithLangChain.py	99-EN/black-myth-wukong/black_myth_wukong_setting.txt	Reads the raw text file using LangChain's TextLoader. Produces a single Document object where page_content contains the entire file text and metadata contains {'source': file_path}.
02-BuildLangChainDocumentObject.py	In-memory string / black_myth_wukong_setting.txt	Manually instantiates Document(page_content=..., metadata={...}). Demonstrates adding custom metadata attributes (author, category, date) required for downstream vector filtering.
03-02-SpecifyParams...LangChain.py	99-EN/black-myth-wukong/*.md	Batch loads all Markdown files in a folder using DirectoryLoader with glob='*.md' and use_multithreading=True. Accelerates ingestion across hundreds of files.
03-03-ChangeTool...LangChain.py	99-EN/black-myth-wukong/	Overrides DirectoryLoader's default Unstructured parser by setting loader_cls=TextLoader. Loads plain text files cleanly without layout analysis overhead.
03-04-SkipErrors...LangChain.py	99-EN/black-myth-wukong/(mixed text & binary folder)	Passes silent_errors=True to DirectoryLoader. Automatically bypasses unreadable, binary, or corrupted files in a directory without crashing the ingestion process.
05-LoadDirectory...LlamaIndex.py	99-EN/black-myth-wukong/	Uses LlamaIndex SimpleDirectoryReader to recursively scan folder files. Automatically selects parsers based on extension, returning LlamaIndex Document objects.
06-LlamaIndex-BuildDocObject.py	In-memory string / custom dict	Constructs LlamaIndex Document objects programmatically, populating extra_info and custom ID keys to customize vector node metadata.
07-UsingUnstructured_v1.py	99-EN/black-myth-wukong/black_myth_wukong_setting.txt	Partitions text into granular structural Element objects (Title, NarrativeText). Inspects element metadata properties via Python's built-in vars() function.
07-UsingUnstructured_v2.py	99-EN/black-myth-wukong/black_myth_wukong_setting.txt	Partitions text into element objects and inspects internal element attributes and category metadata using __dict__ for low-level debugging.

2. Structured Document Loading (02-StructuredDocumentLoading/)

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
01-LangChain-TextLoader-JSON.py	99-EN/black-myth-wukong/black_myth_wukong_characters.json	Ingests JSON files as a raw unparsed text string inside page_content using TextLoader. Fails to leverage key-value structural fields for retrieval.
02-LangChain-JSONLoader-JSON.py	99-EN/black-myth-wukong/black_myth_wukong_characters.json	Uses JSONLoader with jq schema queries (.mainCharacter, .supportCharacters[]). Extracts specific key fields directly into distinct structured Document objects.

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
03-LangChain-WebBaseLoader.py	Wikipedia URL: https://en.wikipedia.org/wiki/Black_Myth:_Wukong	Scrapes live HTML web pages using WebBaseLoader combined with BeautifulSoup SoupStrainer (targeting #bodyContent). Filters out navigation headers, footers, and ads.
04-LangChain-...MarkdownLoader.py	99-EN/black-myth-wukong/ BlackMythWukong-Version...md	Parses Markdown structure (# Headers, lists, blockquotes) using UnstructuredMarkdownLoader into single or multi-element Document objects preserving document hierarchy.
05-01-Unstrutured-SimpleExample.py	Target Web URL / HTML Page	Fetches and categorizes raw web page content into structured visual elements (Header, Title, NarrativeText, ListItem) using Unstructured web parsers.
05-02-Unstrutured-Organize...py	Structured Web HTML / Markdown	Extracts elements and links child paragraphs/lists to parent titles via parent_id metadata attributes, enabling hierarchical parent-child retrieval.

3. Multimodal Image & Presentation Decks (03-ParsingImageAndTextData/)

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
01-Unstructured-ReadImages.py	99-EN/assets/black-myth-wukong/ black_myth_wukong_english.jpg	Extracts embedded textual content from promotional images (.jpg/.png) using Unstructured OCR (Tesseract), creating search-indexable text Documents.
02-Unstructured-ReadPPT.py	90-Data/BlackMythWukong/ black myth wukong.pptx	Partitions PowerPoint presentation decks into slide-by-slide structural elements (Title, NarrativeText, Table), preserving deck sequence.
03-LLM-ReadImagesAndText.py	90-Data/BlackMythWukong/ black myth wukong.pptx / PDF slides	Converts presentation slides to images via pdf2image and passes them to GPT-4o Vision multimodal API. Generates rich text descriptions of complex diagrams and visuals.

4. PDF File Loading & Layout Analysis (04-PDFFileLoading/)

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
01-UsingPyPDF.py	99-EN/black-myth-wukong/ black_myth_wukong_slides.pdf	Standard digital PDF text extraction page-by-page via PyPDFLoader. Fast and lightweight, but struggles with multi-column formats or scanned images.
02-UsingPyMuPDF.py	99-EN/assets/shanxi-tourism/ 山西北上-en.pdf	Uses PyMuPDF (fitz) for high-speed page text extraction, hyperlink collection, and embedded image extraction with high accuracy.
03-UsingPytesseract+pdf2image.py	Scanned Image PDF / Non-searchable PDF	Converts PDF pages into high-DPI images via pdf2image and applies local pytesseract OCR. Recovers text from non-searchable scanned documents.
04-UsingLlamaParser.py	Complex Layout PDF / Travel Guide	Uploads PDF to LlamaParse cloud service. Converts complex multi-column layouts into clean Markdown and parses tables via MarkdownElementNodeParser.
05-LangChain-Unstrutured-PDF...py	99-EN/assets/shanxi-tourism/ 山西北上-en.pdf	Executes UnstructuredPDFLoader with strategy='hi_res' via lazy_load(). Breaks PDF pages into visual elements while retaining original coordinates.

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
06-Unstructured-ParsePDF...v1.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Uses low-level partition_pdf() function to decompose PDF files into raw element objects (Title, NarrativeText, Header, Footer).
06-Unstructured-ParsePDF...v2.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Partitions PDF elements and aggregates statistical counts per element type (e.g., 5 Titles, 42 NarrativeTexts, 3 Tables) for document profiling.
08-RenderPDFPageLayout.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Extracts bounding box coordinates (x1, y1, x2, y2) for each PDF element and renders visual bounding box overlays onto PDF page images for layout inspection.

5. Tabular & Database Data Loading (05-TableDataLoading/)

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
01-01-ImportCSV.py	99-EN/black-myth-wukong/black_myth_wukong.csv	Compares row-by-row table loading using CSVLoader (custom fieldnames & source_column) versus loading the entire file via UnstructuredCSVLoader.
01-02-SpecifyCSVLoader...py	99-EN/black-myth-wukong/*.csv	Configures DirectoryLoader with loader_cls=CSVLoader to batch-ingest all CSV files in a folder into individual row-level Document objects.
02-01-LlamaIndex-SQLDB.py	MySQL Database example_db (game_scenes table)	Connects LlamaIndex DatabaseReader to SQL databases via SQLAlchemy. Executes SQL queries to map database rows directly into Document nodes.
02-02-SQLDB-connection...pymysql.py	MySQL Database example_db	Utility test script verifying low-level socket/TCP database connectivity and table schemas using raw pymysql driver.
02-03-SQLDB-connection...sqlalchemy.py	MySQL Database example_db	Utility test establishing SQLAlchemy connection engine, executing test queries, and converting database tables into pandas DataFrames.
03-01-camelot-ExtractPDFTable.py	90-Data/ComplexPDF/billionaires_page-1-5.pdf	Extracts structured tables from PDF files using camelot-py (Lattice/Stream modes) and exports extracted financial tables directly to CSV or DataFrames.
04-01-pdfplumber-ExtractPDFTable.py	90-Data/ComplexPDF/billionaires_page-1-5.pdf	Parses vector lines and text characters in PDF tables via pdfplumber.extract_tables(), converting complex multi-column rows into structured DataFrames.
04-02-pdfplumber-Extract...QA.py	90-Data/ComplexPDF/billionaires_page-1-5.pdf	Extracts PDF financial tables with pdfplumber, converts table rows to formatted text strings, and indexes them in LlamaIndex for natural language Q&A.
05-03-unstructured-Table...py	90-Data/ComplexPDF/billionaires_page-1-5.pdf	Uses Unstructured partition_pdf(infer_table_structure=True) with layout vision models to extract complex tables into metadata['text_as_html'].
06-01-llamaparser-ExtractPDFTable.py	90-Data/ComplexPDF/billionaires_page-1-5.pdf	Uploads PDF table pages to LlamaParse cloud service. Converts multi-column financial tables into clean Markdown table strings for vector indexing.

6. End-to-End Pipelines & Benchmarks (99-Others/)

Script Filename (01-DataLoading/)	GitHub Dataset Used	What Happens Upon Execution
99-UsingTextract.py	Multi-format files / PDF / Images	Invokes Amazon Textract / textract CLI library to extract raw text content from scanned, proprietary, or non-standard file formats.
15_LangChain_PDF_Unstructured.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Runs an end-to-end RAG benchmark pipeline: loads PDF with UnstructuredPDFLoader, indexes into FAISS vector store, and executes RetrievalQA chain.
15_LlamaIndex_PDF.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Runs an end-to-end RAG benchmark using LlamaIndex PDFReader, constructs VectorStoreIndex, and executes queries via QueryEngine.
15_LlamaParse_PDF_no_Rerank.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Runs an end-to-end benchmark using LlamaParse Markdown output paired with MarkdownElementNodeParser without reranking.
15_LlamaPaser_PDF_Small2Big.py	99-EN/assets/shanxi-tourism/0000-en.pdf	Runs an advanced RAG benchmark using LlamaParse combined with Small-to-Big / Parent-Child retrieval index for superior precision.